



Digital Transformations for
Supporting Next-Generation Labour

Co-funded by the
Erasmus+ Programme
of the European Union



CodeSensei

Nomination I — Virtual university location

Team CodeSensei

Igor Sikorsky Kyiv Polytechnic Institute

Team members

Mark Mamenko, Kateryna Shozda, Denys Ilienکو,
Sviatoslav Pavlenko, Dmytro Poshytyniuk

nextstudy.eu

Mark Mamenko

Igor Sikorsky KPI

Role in the project

Team lead. Secure .NET proxy, Gemini mentor API, Render deployment.

Kateryna Shozda

Igor Sikorsky KPI

Role in the project

Learning design. 24 OOP presets and VR-sized mentor prompts.

Denys Ilienکو

Igor Sikorsky KPI

Role in the project

Quality assurance. NUnit suite, HTTP contract, demo checklist.

Sviatoslav Pavlenko

Igor Sikorsky KPI

Role in the project

VRChat client. UdonSharp terminal, GET-only networking, prefab.

Dmytro Poshytyniuk

Igor Sikorsky KPI

Role in the project

World integration. MetaLab scene, VRChat SDK, placement in the lab.

Authorship

Igor Sikorsky KPI

Role in the project

Created artefacts remain the intellectual property of the team.

Selected Location

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SELECTED LOCATION

MacPaw AI Lab / MetaLab classroom

Igor Sikorsky Kyiv Polytechnic Institute

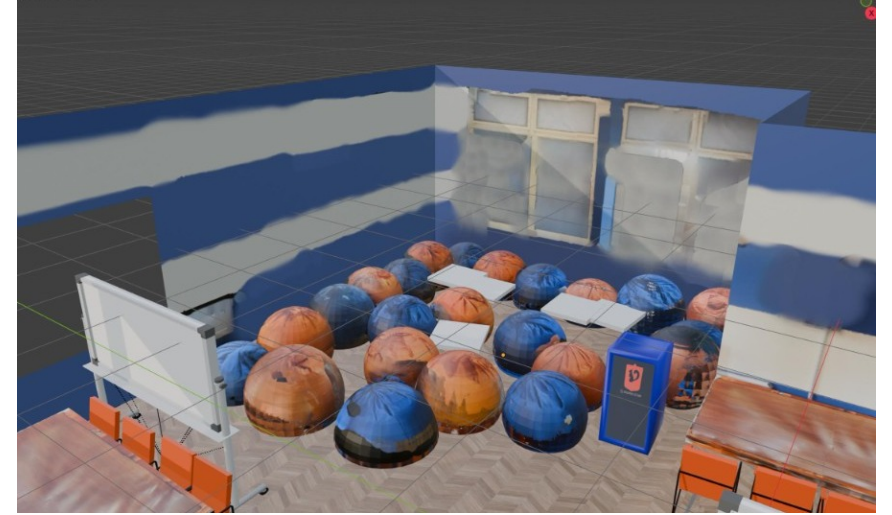
This is the university multimedia classroom modelled for the NEXT Metaverse: blue-and-white walls, herringbone floor, orange seminar seating, whiteboards and a lounge of bean-bags. The MacPaw AI Lab graphic on the end wall marks the space as a live digital-skills studio.

MODELLED SCOPE

- Seminar rows and group tables
- Informal bean-bag lounge
- Whiteboard / critique zone
- CodeSensei terminal at a workstation
- Shared wall for OOP presets
- Circulation to the adjacent bay

Selected Location

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Virtual Model

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Digital Artefacts

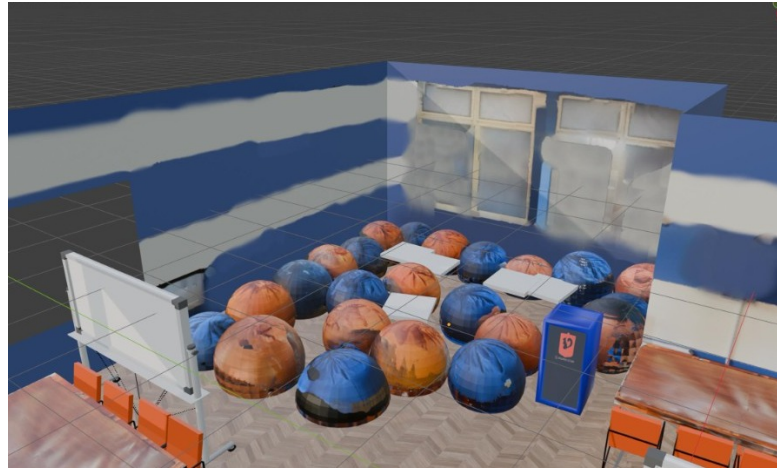
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Classroom model



Full 3D reconstruction of the KPI / MacPaw AI Lab for VRChat.

Informal lounge



Bean-bag zone for informal coaching and peer discussion.

CodeSensei station



Digital artefact: UdonSharp terminal + Gemini mentor in this room.

Technical Implementation

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Tools and software

Unity, VRChat SDK3, UdonSharp, C# / .NET 10, ASP.NET, Docker, Render, Google Gemini, GitHub, NUnit.

Development process

1) Capture the laboratory in 3D. 2) Align a GET-only VRChat contract. 3) Ship the mentor API and 24 OOP presets. 4) Package the terminal prefab for MetaLab.

VRChat integration

VRCStringDownloader only. HTTPS host codesensei-d5zi.onrender.com. Players enable Allow Untrusted URLs. Requests respect the 5.5 s VRChat limit.

Technical challenges

Udon cannot POST, so review uses a 5-character ticket + /paste + inbox. Pending tickets stay hidden. Render Free is woken via /health.

Integration into NEXT-Study Metaverse

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Educational use

In the NEXT-Study Metaverse students sit in this laboratory, open OOP presets or start a live code review. Teachers see the same synced answer on the shared terminal.

User experience

Walk the seminar rows and lounge, press a preset, or generate a ticket, paste a snippet on the web form, and read a mentor reply wrapped to 55 characters for the VR screen.

Further development

Place the prefab at a workstation in this mesh, persist tickets, add languages, and publish the VRChat world after MetaLab import.

Demo and Project Materials

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Project demo

<https://codesensei-d5zi.onrender.com/paste>
<https://codesensei-d5zi.onrender.com/health>

Project materials

GitHub: github.com/mamenkomarkus-creator/CodeSensei-NEXT
Prefab: [client/CodeSensei.unitypackage](#)
Docs: [docs/SUBMISSION.md](#), [docs/api.md](#), [docs/NEXT-INTEGRATION.md](#)

Additional resources

VRChat: hello.vrchat.com NEXT: nextstudy.eu
Authorship of created artefacts remains with the team.



GitHub



Demo



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Thank you!
Questions?

